

MUHAMMAD HAIDER ZAIDI

☎ +49 1523 1698988 ✉ mhaiderzaidi21@gmail.com in [linkedin.com/in/muhammad-haider-zaidi](https://www.linkedin.com/in/muhammad-haider-zaidi)
🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, SQL, Javascript

Data Engineering: PySpark, Airflow, ETL Pipelines, Data Processing, Google Cloud, n8n

Integrations: Onit, Ariba/SAP, Jira, Google Cloud, Slack, DocuSign

AI/ML: Vertex AI, Google Enterprise, BigQuery, AI/ML concepts

Automation: Google Apps Script, Macros, Automation Workflows

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and implemented data pipelines for contract lifecycle management and operational analytics
- Developed scalable data models to support reporting and legal matter management
- Explored novel architectures for AI-based solutions, including Vertex AI and Google Enterprise

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased speed by 5x using AI algorithms
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and time by 6x
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Built an AI chatbot application using the OpenAI API and relevant LLM tools/frameworks
- Full-stack development of a website providing AI-based services using Flask, HTML, CSS

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built web scrapers and ETL pipelines for structured and unstructured data
- Automated data ingestion and cleaning workflows for client analytics use cases

Systems Ltd. - Visionet

Jan 2020 – June 2020

Data Engineer

Lahore, PK

- Gained hands-on experience in Hadoop ecosystem: Hive, Sqoop, NiFi, PySpark, Airflow
- Authored a learning blog on data engineering concepts

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, using AlphaFold for feature extraction and taxonomic bias mitigation

Saaz: AI Self-Playing Chitrali Sitar

2023

Signal Processing, Music Generation

- Designed real-time signal processing system integrating temporal dependencies for music generation
- Implemented transformer-based models for sequence prediction in audio generation tasks

Hugging Face

Open-Source

Contributor

- Co-authored Zero-shot Image Classification documentation and implementation page

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python